EAS439

Examination - January Semester 2025

Numerical Analysis

Tuesday, 15 April 2025 **01:00 pm - 03:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FOUR (4)** questions and comprises **SEVEN (7)** printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book.
4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

EAS439 Copyright © 2025 Singapore University of Social Sciences (SUSS)

Answer all questions. (Full marks: 100)

Question 1

A drone during a vertical landing experiences forces acting on its body as depicted by the free body diagram (FBD) shown in Figure Q1.

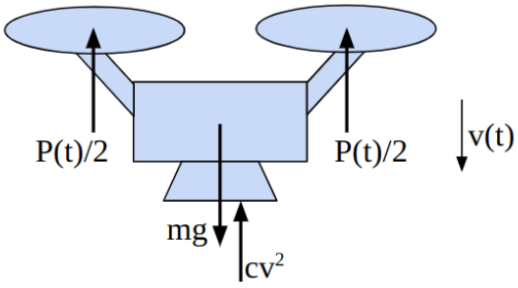

Figure Q1

Question 1a

By applying Newton's 2nd Law, formulate and show that the governing equation for the drone's velocity during landing is

$$\frac{dv}{dt} = g - \frac{c}{m}v^2 - \frac{P(t)}{m}.$$

(5 marks)

Question 1b

Using forward difference (Euler's method), formulate a numerical formula to compute the velocity of the drone at various time-steps during landing. Assume a constant time step of Δt.

(5 marks)

Question 1c

Based on the numerical formula in Question 1(b), calculate the aircraft velocity v(t) using $\Delta t = 0.5$ s. Copy Table Q1(c) to answer the question. The information of the drone's flight is given as follows.

Mass of drone, m = 12 kg.
Drag constant, c = 5 Ns^2/m^2.
Gravitational acceleration, g = 9.81 m/s^2.
Propulsion force, P(t) = 120 N.

Time, t (s)	Velocity, v(t) (m/s)
0	2
0.5	
1.0	
1.5	

Table Q1(c)

(9 marks)

Question 1d

Describe **THREE (3)** methods to appraise the accuracy and validity of the numerical calculations in Question 1c. What can be done to improve the accuracy?

(6 marks)

Question 2a

A system of linear equations is given as

$$\begin{aligned} x + 2y + 3z &= 1, \\ 2x - 6y - z &= 3, \\ x + 0.5y + 0.2z &= 6. \end{aligned}$$

Calculate the solution by applying the Gauss-Seidel iteration method using the initial guess provided below. Copy Table Q2(a) to answer the question (use 4 d.p. for all entries).

Iteration	x	y	z
Initial guess	0	0	0
1			
2			
3			

Table Q2(a)

(11 marks)

Question 2b

The lift coefficient C_L of an airfoil at various angles of attack α (°) can be calculated from the function

$$C_L(\alpha) = 1.35\sin(0.08\alpha) + 0.55\cos(0.05\alpha).$$

Using the Newton-Raphson method, formulate the iterative equation and compute the iterations to find the angle of attack at a lift coefficient of 1.3. Use an initial guess of $\alpha_0 = 0°$. Copy and fill in Table Q2b to answer the question (use 4 d.p. for all entries).

Iteration Number	Root approximation	Relative approx. error (%)
0	0	n.a.
1		
2		
3		

Table Q2b

(11 marks)

Question 3

A 3-meter rod that is insulated at all points except its left end experiences 1D heat conduction along its length as shown in Figure Q3.

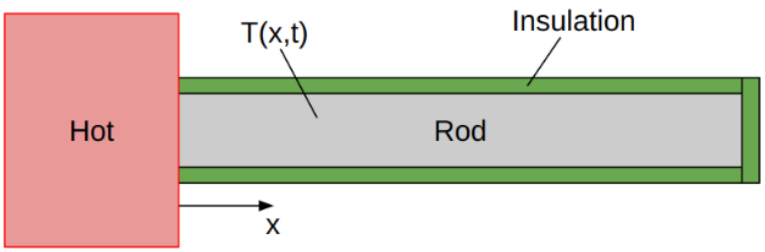

Figure Q3

The temperature $T(x,t)$ along the rod is governed by the 1D heat equation below, where k, ρ and C_p are the thermal conductivity, density and specific heat of the rod, respectively.

$$\frac{\partial T}{\partial t} = \frac{k}{\rho C_p}\frac{\partial^2 T}{\partial x^2}.$$

Question 3a

From the conduction heat flux given by

$$q = -k\frac{\partial T}{\partial x},$$

derive the 1D heat equation as a convective equation in the form below and define the parameter μ.

$$\frac{\partial T}{\partial t} + \mu\frac{\partial q}{\partial x} = 0.$$

(4 marks)

Question 3b

The rod is discretized into 3 finite volumes of equal length as shown in Figure Q3b. Formulate the finite-volume numerical scheme that is explicit in time to solve the rod's temperature at each cell from the convective equation. Assume a constant time step of Δt.

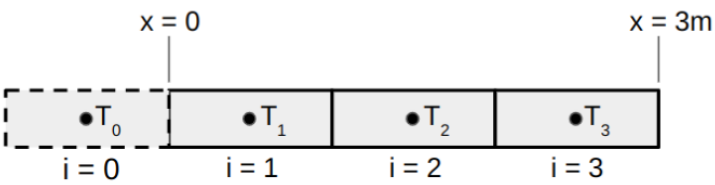

Figure Q3b

(4 marks)

Question 3c

Formulate the heat fluxes at each cell shown in Question 3b. The left end of the rod is maintained at a constant temperature T_0 and the right end is thermally insulated. Copy and fill in Table Q3c to answer this question.

Cell	Left Flux	Right Flux
1		
2		
3		

Table Q3c

(4 marks)

Question 3d

Using the flux formulations in Question 3c and given the temperatures for each cell at t = 1.5 s, compute the left and right fluxes for each cell and their temperatures at t = 2.0 s using $\Delta t = 0.5$ s. Copy and fill in Table Q3d to answer this question. Assume k = 1 and $\mu = 1$.

	T_1	T_2	T_3
t = 1.5 s	62.5 °C	25.0 °C	12.5 °C
Left Flux (W/m^2)			
Right Flux (W/m^2)			
t = 2s			

Table Q3d

(9 marks)

Question 3e

If the 3-meter rod is discretized into 6 segments instead, analyse the maximum Δt that can be applied for the numerical computations in Question 3d to be stable.

(4 marks)

Question 4

The rotor blade of a helicopter experiences the distributed loads as shown in Figure Q4. The blade has a length of 2 m and is approximated as a two-element beam which is cantilevered at node A. Each element has the flexural rigidity as indicated in the figure.

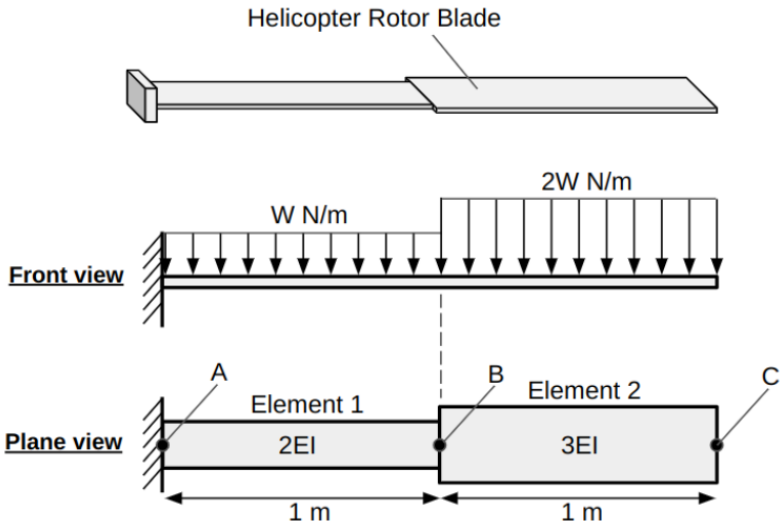

Figure Q4

Each beam element has the matrix equation

$$\begin{bmatrix} f_{1y} \\ m_1 \\ f_{2y} \\ m_2 \end{bmatrix} = \frac{EI}{L^3} \begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix} \begin{bmatrix} v_1 \\ \phi_1 \\ v_2 \\ \phi_2 \end{bmatrix}.$$

Question 4a

Compose the stiffness matrices $k^{(1)}$ and $k^{(2)}$ for each beam element and assemble them to formulate the global stiffness matrix K for the entire rotor blade.

(10 marks)

Question 4b

Analyse the equivalent loads at each node and compose the global loading vector **F**. The reaction force and moment at the fixed end are R and M respectively.

(6 marks)

Question 4c

Apply the boundary conditions and design a numerical scheme to solve for the node displacements and reaction force and moment on the rotor blade. (Describe the procedure and formulate the matrix equations to solve. No actual computations are required.)

(12 marks)

----- END OF PAPER -----